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Are Small LLMs Ready for Coding Agents?

Can tight execution feedback turn structured small-model actions into accepted workflows?

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WHY THIS MATTERS NOW

AskMe gives a small model one structured move at a time

AskMe is an experimental coding-agent harness: explicit plan → one JSON action → execution evidence ↺

01
CONTROLLABLE SMALL LLMS

Choose the model boundary

- Execution speed
- Hardware and deployment
- Post-training access

→

02
ASKME LOOP

One small action at a time

```
plan → act → execute / test
→ fresh evidence ↺
```

Interpret the result. Update locally when possible.

→

03
EXTERNAL ACCEPTANCE LAYER

Check the delivered workflow

- Preserve the full contract
- Check required behavior
- Accept the real artifact independently

The bridge: fixed action vocabulary + fresh execution feedback + external workflow acceptance.

Assumes scoped actions, informative feedback, and independently testable success.

OBSERVED RESULT · QWEN3.6-35B-A3B BUILD CELL

A command passed. The workflow still failed.

REQUIRED CONTRACT

```
write msg.h + main.c
build ./main
run ./main
observe REPLAN_OK
```

A different target was chosen

`cc -o /tmp/test main.c && /tmp/test`

The combined command exited 0

The retained record does not independently prove stdout or source contents.

Agent reported completion

The independent acceptance test found no `./main`.

Exit-zero command + reported completion ≠ accepted artifact.

Current limit: acceptance scored this run after completion; the failure was not returned for recovery.

DESIGN HYPOTHESIS

Hypothesis: update only what fresh evidence invalidates

01
Keep the contract
Carry the required behavior and artifact forward.

→

02
Choose one action
Advance the current plan with bounded scope.

→

03
Execute
Run the tool, focused test, or integration check.

→

04
Interpret
Use the new evidence, not reconstructed state.

→

05
Update locally
Change only what the evidence invalidated.

Expected result
Continue to the next planned action.

Local mismatch
Repair the affected step and rerun its check.

Broken assumption
Replan broadly only when the plan is no longer valid.

Trajectory goal—not yet isolated: fewer repeated failures, fewer stuck steps, and less unnecessary plan churn.

EXPERIMENT 0 · COMPLETED HOSTED SMOKE · JULY 10, 2026

All four variants ran; one deliverable was rejected

4 hosted variants × 2 deliberately simple checks × 1 unseeded trajectory per cell

8 / 8 reported complete

7 / 8 artifact accepted

Compatibility smoke · n=1/cell · descriptive only

MODEL · SHAPE	ARTIFACT BUILD · OBSERVED TRAJECTORY	SANITY REPAIR · OBSERVED TRAJECTORY
Gemma 4 26B A4B ◆ **MoE** · **25.2B total / 3.8B active**	**ACCEPTED** 603.6s · 19 steps · 19.5k tok 1 full / 1 local replan	**ACCEPTED** 20.0s · 7 steps · 4.5k tok 0 full / 0 local replans
Gemma 4 31B ● **Dense** · **30.7B parameters**	**ACCEPTED** 66.5s · 4 steps · 4.4k tok 0 full / 0 local replans	**ACCEPTED** 22.4s · 5 steps · 4.2k tok 0 full / 1 local replan
Qwen3.6-27B ● **Dense** · **27B parameters**	**ACCEPTED** 47.9s · 6 steps · 5.3k tok 0 full / 0 local replans	**ACCEPTED** 23.0s · 6 steps · 4.1k tok 0 full / 0 local replans
Qwen3.6-35B-A3B ◆ **MoE** · **35B total / 3B active**	**NOT ACCEPTED** 17.7s · 3 steps · 3.5k tok wrong artifact path	**ACCEPTED** 11.8s · 4 steps · 2.5k tok 0 full / 0 local replans

Supported
All four reported completion on both checks; acceptance rejected one wrong deliverable.

Not a clean model comparison
No pair isolates size: dense/MoE shape, active compute, run order, and trajectories changed. No Qwen-vs-Gemma, larger-vs-smaller, speed, reasoning, or reliability inference.

Four hosted models × two deliberately simple checks × one unseeded run/cell · evals/README.md

TWO OBSERVED HARNESS BOUNDARIES

Feedback works only when actions enter and failures return

One limit appeared before execution; another after reported completion.

01 · ENTER THE LOOP

The model must emit a valid action

`model → valid structured action`

FeatureBench-fast · 1 task · Gemma 4 31B: 3 plans → 4 reads → 0 writes → empty patch → unresolved.

02 · INSIDE THE LOOP

Execution feedback can guide repair

`action → execute / test → evidence ↺`

AskMe can continue, update locally, or replan.

03 · AFTER COMPLETION

Acceptance checks the deliverable

`artifact → independent acceptance`

The Qwen wrong-path result was rejected, but not returned for recovery.

External boundary probe—not a score: the FeatureBench canary exposed a 512-token structured-action bottleneck before a patch existed. One task cannot separate model capability from this model–harness interface.

Requalified Aug 1 · revision-3 interface · one-task canaries, not scores: sentinel write transport + 4096/8192 budgets + write forcing moved both cells from empty patch to an applied patch — Gemma 4 31B at 11/13 F2P (84.62%), exactly the preregistered pi-ablation ceiling with the identical two failing tests; Qwen3.6-27B at 7/13 (53.85%) vs its 76.92% ceiling. Both still exhausted without `done`; the failure inverted into a commit-without-validate rewrite loop — Gemma rewrote one file 18 times, ran zero tests.

CONCLUSION + LIMITS

Promising for bounded loops. Feature readiness is still open.

Observed

Simple smoke: 7 / 8 artifacts accepted. Feature canary: empty patch → applied at 11/13 F2P, the preregistered ceiling, after the revision-3 interface (one task, changed serving stack — not a score).

Supported

The harness exposed two distinct limits: a wrong deliverable and an action that never reached execution. The revision-3 transport moved that action into execution — and exposed a commit-without-validate rewrite loop.

Still open

Feature-scale reliability beyond one canary; stopping the rewrite loop — validate-after-write pressure, rewrite damping, and an unvalidated-write replan flag are in progress, not merged; reasoning impact; Qwen vs Gemma; size, architecture, and local performance.

Evaluate the model, harness, and task as one system.

Current evidence supports boundary diagnosis—not a general readiness verdict.

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data

A small model's workload depends on the harness

Three technical boundaries—not a leaderboard.

	AskMe	pi	OpenHands
ACTION SURFACE	6 fixed JSON actions; exactly one action per turn.	4 default tools; extensions can add or replace tools.	Typed, extensible `Action → Observation` tools.
STATE + CONTROL	Explicit plan, curated slim state, bounded local or full replanning.	Model-led session tree with branching and lossy compaction; no built-in plan mode.	Conversation state + append-only event log; optional persistence and configurable condenser.
COMPLETION BOUNDARY	`done` → conditional fail-open validation; held-out acceptance external.	Loop ends when tool calls stop; checks come from the workflow or extensions.	`finish` signals completion; benchmark evaluation remains a separate harness.

Trade-off, not ranking: AskMe spends more structure to reduce each turn's decision burden; pi keeps a minimal model-led core; OpenHands supplies a richer lifecycle runtime. All still need independent behavioral acceptance.

Sources: AskMe architecture · pi coding-agent docs · OpenHands tool system and conversation architecture